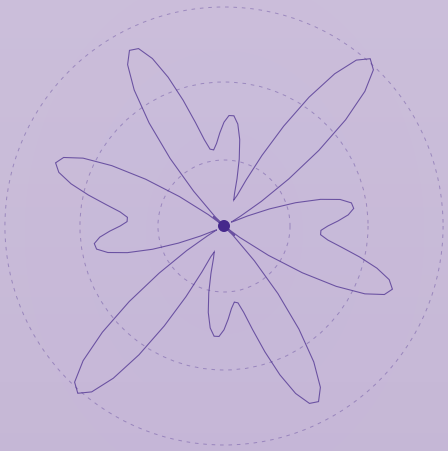
Consciousness



The Subjective Mind and Eleven Neighbors From Global Workspace to Split-Brain Patients

NEUROSCIENCE & PHILOSOPHY

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CONTENTS

— **Introduction** 2
Mapping Subjective Experience Through Its Surrounding Landscape

1. **Phenomenal Versus Access Consciousness** 3
How Philosophy and Neurobiology Split Subjective Experience From Functional Report

2. **Global Neuronal Workspace Ignition** 6
Why a Fifty-Millisecond Word Triggers an All-or-None Prefrontal Broadcast

3. **Integrated Information Theory** 9
Why Sixty-Nine Billion Cerebellar Neurons Contribute Nothing to Subjective Sight

4. **Split-Brain Hemispheric Division** 12
How Severing Two Hundred Million Axons Created Two Independent Streams of Awareness

5. **Blindsight and Cortical Unconscious Routing** 15
How Subcortical Visual Pathways Guide the Hand After Primary Visual Cortex Dies

6. **Binocular Rivalry and Bistable Perception** 18
Where the Primate Visual Hierarchy Switches From Retinal Input to Subjective Percept

7. **Thalamocortical Loops and General Anesthesia** 21
How Propofol Locks the Frontal Cortex and Thalamus Into a Ten-Hertz Cage

8. **Detecting Covert Awareness in Vegetative Patients** 24
How Mental Tennis Inside an MRI Scanner Exposed Consciousness Without Movement

9. **The Perturbational Complexity Index** 27
How Magnetic Cortical Pulses and Lempel-Ziv Compression Measure the Capacity for Experience

10. **Predictive Processing and Controlled Hallucination** 30
How Top-Down Cortical Priors and Precision Weighting Construct the Bodily Self

11. **Higher-Order Metacognition and Prefrontal Monitoring** 33
How Anterior Prefrontal Lesions Blind Introspection While Leaving Vision Intact

12. **The Adversarial Testing of Rival Theories** 36
What Happened When Global Workspace and Integrated Information Theory Faced the Same Data

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Consciousness

Mapping Subjective Experience Through Its Surrounding Landscape

Inside the human skull, eighty-six billion neurons exchange electrochemical action potentials across a hundred trillion synaptic clefts in total physical darkness. No light photons or acoustic pressure waves ever enter the cranial vault. Yet out of those millisecond potassium and sodium fluxes arises an unmistakable inner world. A waking person experiences the crimson hue of a sunset, the sharp sting of a burned finger, and the silent voice of inner speech. Explaining how objective cellular tissue generates a first-person perspective remains the central puzzle of modern mind science.

For most of the twentieth century, behaviorism and early cognitive science tried to sidestep subjective experience altogether. Researchers mapped how sensory receptors transduce energy and how motor cortices command muscle fibers. They treated awareness itself as an untestable philosophical ghost. That evasion collapsed when neurobiologists and clinical neurologists discovered that complex brain computations can run in two distinct modes. A human brain can read flashed words, point accurately at moving lights, mimic fearful faces, and regulate posture without a trace of subjective awareness.

Consciousness becomes tractable only when mapped against its surrounding conceptual neighborhood. Scientists now hold physical stimuli constant while subjective perception flips, or trace what breaks when surgical lesions, magnetic pulses, and general anesthetics alter neural circuits. These methods turn inner experience into a measurable variable. Across this landscape, localized unconscious processing contrasts with global cortical ignition, and feedforward neural chains contrast with integrated cause-effect loops. Even complete bedside paralysis can mask covert mental command-following inside functional neuroimaging scanners.

CHAPTER 1

Phenomenal Versus Access Consciousness

How Philosophy and Neurobiology Split Subjective Experience From Functional Report

At Harvard University in 1960, cognitive psychologist George Sperling seated volunteers in front of a tachistoscope. He flashed a rectangular three-by-four grid of twelve consonant letters for exactly fifty milliseconds. Once the grid vanished into a blank white field, subjects had to write down every letter they had seen. Across hundreds of trials, observers hit a stubborn ceiling. They could report on average only 4.5 letters out of the twelve before the memory trace faded. Yet every participant insisted that at the moment of the flash they had clearly seen all twelve letters at once.

To test whether the subjects suffered from an illusion or actually saw more than they could say, Sperling introduced an acoustic partial-report cue. Immediately after the fifty-millisecond visual flash ended, a loudspeaker sounded a high, medium, or low musical tone. The pitch instructed the observer to report only the top, middle, or bottom row of four letters. Because the tone arrived *after* the visual display had turned off, the observer's eyes could not focus on one row in advance. Nevertheless, subjects accurately reported more than three of the four letters in whichever row the tone selected. Roughly nine to ten letters remained visibly present in short-lived sensory memory, even though only four could pass through the narrow bottleneck of verbal report.

Thirty-five years later, in 1995, philosophers David Chalmers and Ned Block independently published landmark papers that used experiments like Sperling's to draw the map of modern consciousness research. Speaking first at the 1994 Tucson conference at the

University of Arizona and publishing the following year in the *Journal of Consciousness Studies*, Chalmers separated the scientific study of mind into two distinct categories. That same year in *Behavioral and Brain Sciences*, Block divided conscious states along a parallel line. Both philosophers built on the empirical program launched in 1990 by molecular biologist Francis Crick and physicist Christof Koch:

- **The Easy Problems and Access Consciousness (A-Consciousness)**: Explains how neural circuits discriminate sensory stimuli and integrate information across brain regions. It covers how representations enter working memory to guide deliberate behavior and verbal report.
- **The Hard Problem and Phenomenal Consciousness (P-Consciousness)**: Confronts why those functional computations feel like something from the inside. It asks why wavelength discrimination is accompanied by the felt quality of redness rather than running in the dark like an automated thermostat.
- **The Neural Correlates of Consciousness (NCC)**: Defines the empirical target proposed by Crick and Koch in *Seminars in the Neurosciences* in 1990. It isolates the minimal set of neuronal events and mechanisms jointly sufficient for a specific conscious percept.

Block argued that Sperling's twelve-letter grid exposes a real split between phenomenal experience and cognitive access. During the brief flash, observers possess phenomenal consciousness of the entire twelve-letter array. Access consciousness, by contrast, is restricted to the four items routed into fronto-parietal working memory for verbal report. Critics counter that observers merely experience a coarse summary statistic of the un-cued rows until the tone directs attention to convert one row into detailed access. Regardless of how that debate resolves, the division between phe-

nomenal experience, cognitive access, and minimal neural correlates transformed consciousness into a set of laboratory protocols. When Sperling delayed the acoustic cue by just one full second after the twelve-letter display vanished from his Harvard tachistoscope in 1960, partial-report performance collapsed right back to the 4.5-letter ceiling of un-cued whole report.

CHAPTER 2

Global Neuronal Workspace Ignition

Why a Fifty-Millisecond Word Triggers an All-or-None Prefrontal Broadcast

Inside the neuroimaging unit of the Service Hospitalier Frédéric Joliot in Orsay, France, in 2001, cognitive neuroscientist Stanislas Dehaene and neurologist Lionel Naccache scanned healthy volunteers with functional magnetic resonance imaging and sixty-four-channel scalp electroencephalography. On a screen before the subjects, four-letter French nouns flashed for just twenty-nine milliseconds. Immediately before and after each word appeared, the screen displayed a dense geometric pattern of overlapping diamonds and pound signs. That forward-and-backward visual masking completely erased the words from subjective awareness. Subjects reported seeing only flickering geometric shapes and performed at chance when asked whether a word had appeared.

When Dehaene and Naccache inspected the brain scans from those subliminal twenty-nine-millisecond trials, the visual system had not ignored the hidden words. Feedforward retinal signals traveled through the primary visual cortex and activated the left fusiform gyrus. This extrastriate region, known as the visual word-form area, extracts letter strings regardless of whether a word is printed in uppercase or lowercase. Yet this localized sensory wave died out within two hundred milliseconds. It never reached the front of the brain.

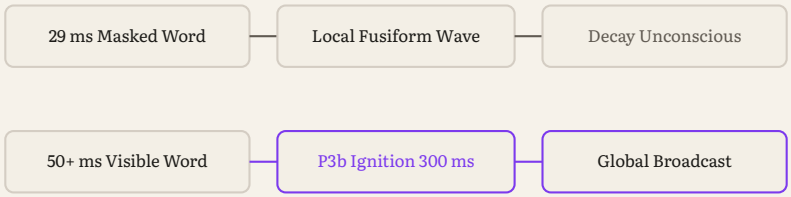


FIG. 2.1 — Subliminal local decay below 40 ms (top) versus nonlinear fronto-parietal workspace ignition above 50 ms (bottom).

To explain what separates a dying local wave from a reportable conscious percept, Dehaene and neurobiologist Jean-Pierre Changeux at the Collège de France built a neural model of psychologist Bernard Baars's 1988 Global Workspace theory. Baars had compared the human mind to a theater stage. Dozens of specialized, unconscious neural modules process syntax, motion, faces, and motor plans in parallel in the dark audience. Consciousness corresponds to the narrow spotlight on the central stage, whose contents are broadcast back to every processor in the house. Dehaene and Changeux identified the physical wiring of that broadcast stage in a dense web of long-range excitatory pyramidal neurons. Concentrated in cortical layers II and III, these long-axon cells reciprocally link the dorsolateral prefrontal cortex, anterior cingulate, and inferior parietal lobules.

When a sensory signal is weak or cut off by a backward mask within forty milliseconds, bottom-up excitation fails to overcome local inhibitory interneurons. The signal remains trapped inside modular sensory processors. Even while trapped locally, these subliminal traces are not completely inert. In behavioral priming experiments at Orsay, a masked twenty-nine-millisecond word still sped up a volunteer's subsequent button press when classify-

ing a related target word by twenty to thirty milliseconds. Yet the volunteer could not name the prime, hold it in working memory for two seconds, or use it to guide a deliberate verbal choice. Once stimulus duration crosses a threshold around fifty milliseconds, however, top-down recurrent excitation from prefrontal and parietal hubs amplifies the sensory trace in a self-sustaining loop. In dynamical systems terms, the network crosses a nonlinear bifurcation—an all-or-none phase transition called global ignition.

In 2005, Claire Sergent, Sid Kouider, and Stanislas Dehaene tracked this exact bifurcation in human volunteers using continuous visibility ratings and high-density electroencephalography. During the first 180 milliseconds after stimulus onset, visual cortex potentials rose linearly with stimulus strength whether or not the subject noticed the word. Around 270 to 300 milliseconds post-stimulus, the neural trajectories split into two distinct modes. Unseen stimuli collapsed back to baseline. Consciously seen stimuli triggered a massive late positive voltage wave—the P3b component—accompanied by long-range beta and gamma synchrony across fronto-parietal electrodes. In Dehaene and Naccache's 2001 neuroimaging trials at Orsay, removing the pattern masks so the twenty-nine-millisecond words became consciously visible boosted metabolic blood-oxygen-level-dependent activity in the left fusiform gyrus twelvefold. Simultaneously, that unmasked broadcast ignited newly recruited prefrontal and parietal regions across 24,496 cubic millimeters of cortex, compared with just 2,176 cubic millimeters during masked subliminal trials.

CHAPTER 3

Integrated Information Theory

Why Sixty-Nine Billion Cerebellar Neurons Contribute Nothing to Subjective Sight

Count every neuron inside the adult human brain and a striking anatomical asymmetry emerges. In 2009, neuroscientist Suzana Herculano-Houzel and her colleagues at the Federal University of Rio de Janeiro dissolved whole post-mortem human brains into homogeneous nuclear suspensions to perform exact isotropic fractionator cell counts. They proved that the adult human brain contains an average of 86.1 billion neurons. Fully 69.0 billion of those neurons—roughly eighty percent of the entire brain's cellular total—are packed inside the fist-sized cerebellum at the back of the skull. The entire cerebral cortex, by contrast, holds only 16.3 billion neurons.

Yet if a neurosurgeon removes a patient's cerebellum due to a tumor, or if a person is born without a cerebellum altogether, their conscious experience of the world remains untouched. Such patients suffer from slurred speech, unsteady gait, and jerky limb movements. Still, they see colors, hear music, feel grief, and converse with undiminished subjective richness. Conversely, a small stroke or localized pharmacological suppression in the thalamocortical network—containing fewer than a quarter as many neurons—can extinguish conscious awareness completely.

Why do sixteen billion cortical neurons generate an inner world while sixty-nine billion cerebellar neurons operate in total experiential darkness? In 2004, psychiatrist and neuroscientist Giulio Tononi at the University of Wisconsin–Madison proposed Integrated Information Theory (IIT) to solve this exact anatomical

puzzle. Rather than asking what a neural circuit does for external behavior, Tononi started from essential properties of subjective experience itself. Every conscious state is intrinsic to the subject, unified into a single whole that cannot be subdivided into independent parts, and highly informative because it rules out billions of alternative experiential states.

To translate these properties into mathematical physics, Tononi analyzed a network of interacting elements with state vector X_t governed by a causal transition probability matrix. A system generates conscious experience only to the extent that its elements constrain their own past and future states *more* as an integrated whole than they do when cleaved across their weakest causal link. Tononi named that weakest cut the Minimum Information Partition ($\mathcal{P}_{\mathrm{MIP}}$). In Tononi's foundational formulation, integrated information Φ measures the informational divergence D between the unpartitioned network's cause-effect repertoire and the product of its severed parts $M^{(k)}$ across that weakest cut:

$$\Phi = \min_{\mathcal{P}} D\Big(p\big(X_t \mid X_{t-1}\big) \,\Big\|\, \prod_{k \in \mathcal{P}} p\big(M_t^{(k)} \mid M_{t-1}^{(k)}\big)\Big)$$

Look at what happens to Φ inside the cerebellum. Cerebellar granule cells and Purkinje cells are wired into millions of parallel, non-overlapping feedforward microzones. Mossy fibers feed granule cells, which project to Purkinje cells and exit through deep cerebellar nuclei to refine motor timing. Almost no lateral excitatory feedback connects one cerebellar microzone to its neighbors. Because independent parallel channels do not causally constrain one another, slicing the cerebellum between microzones leaves each module's transition probabilities unchanged. The partitioned product equals the whole, driving Φ near zero across each cut despite the sixty-nine billion cells.

The thalamocortical system, by contrast, combines high functional specialization with dense recurrent horizontal and vertical feedback loops. Cutting the cerebral cortex across any boundary severs thousands of reciprocal causal links, yielding a high Φ value for the main complex. When neurosurgeons at the General Hospital of Jinan Military Area Command scanned a female patient in 2014 who had walked and conversed for more than two decades despite mild gait ataxia, magnetic resonance imaging revealed an empty posterior cranial fossa filled with cerebrospinal fluid where all sixty-nine billion cerebellar neurons were congenitally absent, without diminishing her conscious perceptual life.

CHAPTER 4

Split-Brain Hemispheric Division

How Severing Two Hundred Million Axons Created Two Independent Streams of Awareness

Two hundred million myelinated nerve fibers form the corpus callosum, the thick white-matter bridge connecting the left and right cerebral hemispheres of the human brain. In 1962, Los Angeles neurosurgeons Philip Vogel and Joseph Bogen severed the entire corpus callosum and anterior commissure of a World War II paratrooper known by the initials W.J. The patient had suffered from intractable grand mal epileptic seizures for fifteen years after wartime head trauma. The operation halted his generalized convulsions. In casual conversation and standard bedside neurological exams, W.J. appeared completely unchanged in personality, humor, and intellect.

At the California Institute of Technology, however, neurobiologist Roger Sperry and his graduate student Michael Gazzaniga built a divided-field tachistoscope to test W.J.'s two hemispheres separately. Because nasal retinal fibers cross at the optic chiasm, every visual stimulus located to the left of a central fixation point projects exclusively to the right visual cortex. Every stimulus to the right of fixation projects exclusively to the left visual cortex. In a healthy person, the corpus callosum shares both halves of the visual scene across the midline in milliseconds. And if an image stays on a screen for more than two hundred milliseconds, a split-brain patient simply flicks their eyes in a rapid saccade so both hemispheres see it. Sperry and Gazzaniga defeated eye movements by flashing images for just one hundred milliseconds—faster than the oculomotor system can initiate a saccade.



FIG. 4.1 — Tachistoscopic split-brain routing: right visual field to verbal left cortex (top) vs left visual field to mute right cortex (bottom).

When Gazzaniga flashed a photograph of a spoon to W.J.'s right visual field, the signal entered the left hemisphere. Because the left hemisphere houses Broca's and Wernicke's speech centers in over ninety-five percent of right-handed people, W.J. immediately said he saw a spoon. When a picture of a key was flashed to his left visual field, however, the signal entered only the disconnected right hemisphere. Asked what he had seen, W.J.'s speaking left hemisphere sincerely reported seeing nothing at all, or merely an empty white flash. Yet when Gazzaniga asked W.J. to reach beneath an opaque curtain with his left hand—controlled by the right hemisphere's motor cortex—and select the matching object by touch, his left hand bypassed the spoon and picked up the key every time. Asked what his left hand was holding behind the screen, W.J.'s speaking hemisphere had to guess at random.

Even more revealing were the simultaneous double-picture experiments conducted in the late 1970s by Gazzaniga and neuroscientist Joseph LeDoux on commissurotomy patient P.S. Unlike W.J., P.S. had a right hemisphere with enough receptive vocabulary to comprehend written nouns despite being unable to speak. On one trial, the researchers flashed a picture of a chicken claw to P.S.'s left hemisphere while simultaneously flashing a winter snow

scene to his right hemisphere. Placed before an array of choice cards, P.S.'s right hand pointed to a chicken while his left hand pointed to a snow shovel.

Asked why his two hands had chosen two unrelated items, P.S. never admitted that he did not know why his left hand picked the shovel. Without hesitation, his verbal left hemisphere—which had seen only the chicken claw and watched the left hand touch the shovel—explained that the chicken claw went with the chicken, and a person needs a shovel to clean out the chicken shed. Gazzaniga named this left-hemisphere system the interpreter: a narrative module that continuously observes the body's actions and fabricates coherent post-hoc explanations for behaviors initiated outside its own conscious awareness. When Gazzaniga and LeDoux flashed the written command to walk exclusively to the left visual field of patient P.S. in 1977, his mute right hemisphere read the word and caused him to stand up and walk toward the testing-room door, whereupon his verbal left hemisphere—asked why he had risen—immediately claimed that he was going to get a cold drink.

CHAPTER 5

Blindsight and Cortical Unconscious Routing

How Subcortical Visual Pathways Guide the Hand After Primary Visual Cortex Dies

Clinical intake logs at the National Hospital for Nervous Diseases at Queen Square, London, in 1973 recorded a stark visual deficit in a male patient known as D.B. To halt severe migraines and visual obscurations caused by a benign arteriovenous malformation at the pole of his right occipital lobe, neurosurgeons excised the malformation along with his right primary visual cortex (striate cortex, area V1). Standard perimetry mapping after surgery confirmed dense homonymous hemianopia. D.B. was clinically blind across almost the entire left half of his visual field.

When Oxford psychologist Lawrence Weiskrantz and neuropsychologist Elizabeth Warrington tested D.B. in 1974, they began by holding targets inside his blind left hemifield and asking what he saw. D.B. insisted that he saw nothing whatever—neither shape, nor flash, nor movement. Instead of accepting his verbal report and ending the exam, Weiskrantz changed the protocol to a two-alternative forced-choice task. He asked D.B. to reach out and touch the spot on a wall screen where a brief light had flashed inside his blind field. In another set of trials, D.B. had to guess whether an unseen plaque carried horizontal or vertical stripes.

To D.B.'s own disbelief, when he stopped trying to see the target and simply guessed or let his hand reach automatically, his finger landed squarely on the invisible light coordinates. His orientation guesses exceeded ninety percent accuracy across hundreds of tri-

als. Weiskrantz coined the term *blindsight* in a landmark 1974 *Brain* paper to describe this preservation of accurate visual discrimination inside a cortical scotoma devoid of phenomenal awareness. Over the next three decades, anatomical tracing and functional neuroimaging mapped the three parallel visual streams that explain why destroying V1 abolishes conscious sight while sparing unconscious action and emotion:

- **The Conscious Geniculostriate Pathway (Retina → LGN → V1)**: Carries roughly ninety percent of retinal ganglion cell axons through the dorsal lateral geniculate nucleus of the thalamus into primary visual cortex V1, feeding the ventral temporal stream required for conscious visual perception.
- **The Unconscious Retinotectal Action Pathway (Retina → Superior Colliculus → Pulvinar → MT/V5)**: Routes the remaining ten percent of retinal fibers through the midbrain superior colliculus and thalamic pulvinar directly to dorsal motion area MT/V5 and posterior parietal cortex, guiding hand reaching and eye saccades without V1.
- **The Subcortical Affective Pathway (Superior Colliculus → Pulvinar → Amygdala)**: Delivers coarse, low-spatial-frequency visual signals of biological threat directly to the limbic amygdala, triggering autonomic arousal and facial muscle mimicry before cortical object recognition occurs.

This third pathway produced an even stranger clinical dissociation in 1999. Cognitive neuroscientist Beatrice de Gelder and Lawrence Weiskrantz tested patient G.Y., who had lost his left primary visual cortex in a childhood road accident. De Gelder projected video clips of actresses making happy or fearful facial expressions strictly inside G.Y.‘s blind right hemifield. Although G.Y. reported zero conscious awareness of seeing any face, he correctly guessed the emotional expression far above chance.

Functional MRI later confirmed robust bilateral activation in his amygdala via the superior colliculus and pulvinar. In Beatrice de Gelder's 1999 affective blindsight recordings on patient G.Y. at Tilburg University and Oxford, bipolar electromyography electrodes taped over his cheek and brow detected congruent smiling or frowning muscle contractions within 300 to 500 milliseconds after a happy or fearful face appeared inside his blind hemifield—faster than the voluntary facial responses elicited in his intact sighted field.

CHAPTER 6

Binocular Rivalry and Bistable Perception

Where the Primate Visual Hierarchy Switches From Retinal Input to Subjective Percept

Place a vertical mirror reflection of the letter A before your left eye and a reflection of the letter S before your right eye using the mirror stereoscope invented by physicist Charles Wheatstone at King's College London in 1838. Because the two contours fall on corresponding retinal coordinates but cannot be fused into a single three-dimensional object, the visual system faces an irreconcilable conflict. Instead of perceiving a superimposed transparent jumble of both letters, your conscious sight alternates spontaneously every two to four seconds. First the A fills awareness while the S vanishes completely. Moments later, a wave of dominance sweeps across the visual field, erasing the A and revealing the S.

For consciousness researchers in the 1990s, Wheatstone's binocular rivalry offered an ideal experimental scalpel. Throughout a rivalry session, the physical photons striking both retinas remain 100 percent constant, yet the observer's conscious percept flips back and forth. Suppose a neurophysiologist records from single neurons along the primate visual pathway while a subject reports these alternations. Which brain areas loyally track the constant physical stimulus on the retina, and which areas flip in lockstep with subjective awareness?

Between 1989 and 1997, neuroscientist Nikos Logothetis and his colleagues Jeffrey Schall, David Leopold, and David Sheinberg at Baylor College of Medicine answered that question in awake behaving rhesus macaques (*Macaca mulatta*). The monkeys were trained to pull left or right levers to report whether they perceived

leftward-tilted gratings, rightward-tilted gratings, sunburst patterns, or faces presented separately to the two eyes. Using tungsten microelectrodes, Logothetis's team recorded extracellular action potentials from hundreds of individual neurons across four successive rungs of the macaque visual cortex.

The recordings revealed a steep hierarchical gradient across the cortex. In primary visual cortex (area V1), roughly eighty-two percent of orientation-selective neurons kept firing steadily to whichever grating hit their preferred eye. They remained completely indifferent to whether the monkey consciously saw that grating or the rival one; only eighteen percent modulated their firing rate with the monkey's reported percept. One rung higher, in intermediate visual areas V4 and MT, the fraction of percept-tracking neurons rose to roughly forty percent.

To model how competing cortical pools generate spontaneous dominance switches from steady retinal inputs I_1 and I_2, computational neuroscientists Hugh Wilson and Carson Chow formulated coupled rate equations. These equations govern the excitatory firing rates E_1, E_2 and slow spike-frequency adaptation currents A_1, A_2:

$$\tau \frac{dE_1}{dt} = -E_1 + \sigma\big(I_1 - \beta E_2 - g A_1\big), \qquad \tau_A \frac{dA_1}{dt} = -A_1 + E_1$$

Here, the dominant neuronal pool E_1 suppresses its rival E_2 through GABAergic cross-inhibition of strength β. While E_1 fires rapidly, however, slow calcium-dependent potassium channels build up the hyperpolarizing adaptation variable A_1 over a long time constant $\tau_A \gg \tau$. Once self-adaptation $g A_1$ weakens the dominant pool below the cross-inhibition threshold, the suppressed pool E_2 escapes inhibition and takes over awareness. This dynam-

ical balance also explains the counterintuitive psychophysical law discovered by Dutch psychologist Willem Levelt in 1965. Increasing the physical contrast of the image shown to one eye does not lengthen how long that strong eye stays dominant. Instead, higher contrast shortens the dominance duration of the *other* eye by overcoming its inhibitory adaptation faster. At the top of the ventral visual stream, in the inferior temporal (IT) cortex and superior temporal sulcus, Sheinberg and Logothetis found in 1997 that ninety percent of responsive neurons fired vigorously when their preferred face or sunburst pattern dominated awareness. Those same inferior temporal neurons dropped to baseline firing rates the instant the rival image suppressed the preferred stimulus, even though both images remained focused on the retinas.

CHAPTER 7

Thalamocortical Loops and General Anesthesia

How Propofol Locks the Frontal Cortex and Thalamus Into a Ten-Hertz Cage

Every day in surgical operating rooms around the world, anesthesiologists inject two milligrams per kilogram of propofol into a patient's vein. Discovered by veterinarian John Glen at Imperial Chemical Industries in 1977, propofol is a milky intravenous emulsion of 2,6-diisopropylphenol. Within forty seconds of injection, the patient stops counting backward, loses their eyelash reflex, and enters a state of reversible coma. Even a surgical scalpel incision evokes zero conscious pain or memory. For decades, textbook physiology assumed that general anesthetics extinguish consciousness simply by depressing neuronal firing across the brain until the cerebral cortex falls silent.

When anesthesiologist and statistician Emery Brown at Massachusetts General Hospital and the Massachusetts Institute of Technology analyzed high-density electroencephalograms (EEG) during propofol induction between 2010 and 2013, he encountered a striking paradox. At the exact moment a patient loses consciousness, electrical power over the frontal scalp does not flatten toward zero. Instead, the low-amplitude, high-frequency gamma and beta waves of waking thought are replaced by massive, highly regular ten-hertz alpha oscillations paired with slow one-hertz delta waves. Known clinically as anteriorization, this shift migrates alpha power from the back of the head to the forehead.

The anesthetized frontal cortex actually fires in louder, more synchronized rhythmic lockstep than the waking brain.

How does amplifying a regular ten-hertz rhythm erase subjective awareness? Working with computational neuroscientist Nancy Kopell and neurophysiologist Patrick Purdon, Brown traced propofol's circuit mechanism from single ion channels up to the thalamocortical loop. At the molecular level, propofol binds to post-synaptic $\mathrm{GABA_A}$ (gamma-aminobutyric acid) receptor chloride channels on inhibitory interneurons across the cerebral cortex and inside the shell-like thalamic reticular nucleus. By slowing the unbinding of GABA from its receptor pocket, propofol doubles or triples the decay time constant τ_{decay} of the inhibitory synaptic gating variable $s(t)$ governing the hyperpolarizing chloride current I_{GABA}:

$$I_{\mathrm{GABA}}(t) = g_{\mathrm{GABA}}\, s(t)\,\big(V(t) - E_{\mathrm{Cl}}\big), \qquad \frac{ds}{dt} = \alpha_s[\mathrm{GABA}](1 - s) - \frac{s}{\tau_{\mathrm{decay}}}$$

Because the chloride reversal potential $E_{\mathrm{Cl}} \approx -75\ \mathrm{mV}$ sits below the resting membrane voltage $V(t)$, this prolonged chloride influx hyperpolarizes thalamocortical relay neurons. In thalamic relay cells, sustained hyperpolarization removes inactivation from specialized low-threshold T-type calcium channels (I_T). When the prolonged $\mathrm{GABA_A}$ inhibition finally wears off after roughly fifty milliseconds, the thalamic relay cells fire a rebound calcium spike and a burst of action potentials up to the frontal cortex. Those cortical pyramidal cells immediately excite the thalamic reticular nucleus to re-inhibit the thalamus for another fifty-millisecond half-cycle.

That fifty-millisecond inhibition plus fifty-millisecond rebound creates a self-sustaining one-hundred-millisecond period—an ex-

act ten-hertz thalamocortical oscillation. Rather than shutting neurons off, propofol traps the thalamus and frontal cortex inside a narrow ten-hertz rhythmic cage. Locked into synchronous oscillation, the circuit retains zero bandwidth for the differentiated spike patterns required to broadcast sensory information across the brain. Simultaneously, propofol suppresses brainstem arousal nuclei that normally depolarize the thalamus with acetylcholine. When the intravenous propofol infusion is turned off and the drug clears from $GABA_A$ receptors, the ten-hertz frontal alpha rhythm dissolves. Normal desynchronized gamma activity resumes and consciousness returns within minutes. In 2012, Laura Lewis, Emery Brown, and Sydney Cash recorded simultaneously from eighty-four single cortical neurons in human neurosurgical patients during propofol induction at Massachusetts General Hospital. They found that the exact second the patients stopped responding to auditory clicks coincided with abrupt 0.5-hertz slow oscillations. Those slow waves silenced ninety-five percent of neuronal spiking during every DOWN phase and broke phase coherence between cortical sites separated by more than two centimeters.

CHAPTER 8

Detecting Covert Awareness in Vegetative Patients

How Mental Tennis Inside an MRI Scanner Exposed Consciousness Without Movement

Five months after suffering severe traumatic brain injury in a July 2005 road traffic accident, a young female patient entered Addenbrooke's Hospital in Cambridge, England, for multidisciplinary neurological assessment. She opened her eyes spontaneously, maintained regular sleep-wake cycles, and breathed without mechanical assistance. Yet across repeated standardized bedside exams, she showed zero voluntary movement, zero visual tracking of a moving mirror, and zero verbal or gestural response to spoken commands. Under every international neurological criterion, she sat in an unambiguous vegetative state. That diagnosis assumed that while her brainstem remained alive, her cerebral cortex had lost all capacity for conscious awareness.

Cognitive neuroscientist Adrian Owen at the Medical Research Council Cognition and Brain Sciences Unit in Cambridge and neurologist Steven Laureys at the University of Liège suspected that bedside behavioral exams suffered from a fatal flaw. Every clinical test of consciousness requires the patient to move a muscle—to squeeze a hand, blink an eye, or speak a word. Suppose diffuse axonal injury or subcortical lesions sever the motor output pathways from cortex to spinal cord while sparing the thalamocortical network itself. In that case, an awake, thinking mind could lie entombed inside a paralyzed body.

In September 2006, Owen and Laureys published a landmark study in *Science* that bypassed the muscular system altogether. They slid the unresponsive Cambridge patient into a three-Tesla functional magnetic resonance imaging (fMRI) scanner. Through headphones, they delivered spoken instructions across alternating thirty-second blocks and compared her cortical blood-oxygen-level-dependent (BOLD) responses against twelve healthy volunteers:

- **Motor Imagery Task (Playing Tennis)**: Instructed the patient to imagine swinging a racket back and forth in a vigorous game of tennis for thirty seconds. This task reliably activates the supplementary motor area (SMA) in healthy human brains.
- **Spatial Navigation Task (Walking Through Home)**: Instructed the patient to imagine walking from room to room through her house starting from the front door. This task selectively activates the parahippocampal gyrus, posterior parietal lobe, and lateral premotor cortex.
- **Binary Yes/No Communication Protocol (Monti et al., 2010)**: Assigned tennis imagery (SMA activation) to mean *yes* and house-walking imagery (parahippocampal activation) to mean *no*. This mapping allowed behaviorally unresponsive patients to answer autobiographical questions inside the scanner.

When Owen and Laureys analyzed the Cambridge patient's scans, her supplementary motor area lit up promptly at the start of every thirty-second tennis block and subsided at the rest cue. Conversely, her parahippocampal gyrus and parietal cortex lit up throughout every thirty-second house-navigation block. Because the hemodynamic response persisted for the full thirty seconds of each trial before switching on command, the activations could not be dismissed as automatic single-word acoustic reflexes. Four years later, in a 2010 *New England Journal of Medicine* study of fifty-

four vegetative and minimally conscious patients in Cambridge and Liège led by Martin Monti, Owen, and Laureys, five patients willfully modulated their brain activity on command. One traumatic brain injury patient used the tennis-versus-navigation code to answer five of six autobiographical questions accurately, including confirming his father's name was Alexander. Neurologist Nicholas Schiff formalized this condition in 2015 as cognitive motor dissociation. In August 2024, Yelena Bodien, Nicholas Schiff, Steven Laureys, Adrian Owen, and the Coma Recovery Consortium analyzed fMRI and EEG recordings from 241 behaviorally unresponsive brain-injured patients across six medical centers. Sixty of those patients—fully twenty-five percent of the cohort—reliably activated motor or spatial cortical networks on verbal command despite failing bedside behavioral subscales.

CHAPTER 9

The Perturbational Complexity Index

How Magnetic Cortical Pulses and Lempel-Ziv Compression Measure the Capacity for Experience

Functional MRI mental-imagery tasks revolutionized coma neurology, yet they carry a steep clinical limitation. To imagine playing tennis for thirty seconds inside a noisy magnet, a brain-injured patient must possess intact hearing, intact language comprehension, sustained working memory, and executive motivation. A patient suffering from aphasia, Wernicke's deafness, or fluctuating delirium will fail the tennis test even when vividly conscious. So will a healthy sleeper immersed in a rapid-eye-movement (REM) dream who ignores external sounds. Neurologists needed an objective meter of consciousness that required zero sensory input and zero voluntary cooperation from the patient.

At the University of Wisconsin–Madison in 2005 and the University of Milan in 2013, neurophysiologist Marcello Massimini and Giulio Tononi built that meter by adapting the oldest method in physical inspection. They struck the cerebral cortex with a brief perturbation and listened to the complexity of its electrical echo. They called their bedside protocol *zap and zip*. First, the investigator holds a figure-eight Transcranial Magnetic Stimulation (TMS) coil against the patient's scalp and delivers a focal, painless magnetic pulse. That magnetic field directly depolarizes a patch of cortical neurons, bypassing ears, eyes, and subcortical sensory nerves. Second, a sixty-channel high-density EEG cap records the resulting electrical echo across the cerebral cortex over the next three hundred milliseconds.

Using inverse source modeling and nonparametric bootstrap statistics, Massimini and physicist Adenauer Casali converted the evoked cortical currents into a binary spatiotemporal matrix $\mathbf{S}(x,t)$ of total length L. Each entry equals 1 if cortical source x shows statistically significant activation at millisecond t and 0 otherwise. They then compressed that binary string using the 1976 Lempel-Ziv algorithmic complexity algorithm $c_L(\mathbf{S})$—the same dictionary-building principle behind `.zip` computer files—and normalized it by the source Shannon entropy $H(p_1)$ of the active fraction p_1 to define the Perturbational Complexity Index (PCI):

$$\mathrm{PCI} = \frac{c_L(\mathbf{S}) \cdot \log_2 L}{L \cdot H(p_1)}, \qquad H(p_1) = -p_1 \log_2 p_1 - (1-p_1)\log_2(1-p_1)$$

Why does Lempel-Ziv compression of a magnetic echo isolate consciousness? According to Integrated Information Theory, a conscious brain must simultaneously possess high *integration* (long-range causal interaction) and high *differentiation* (a large repertoire of distinct local states). During deep non-REM slow-wave sleep, cortical effective connectivity breaks down. A TMS pulse evokes a brief local voltage bump right beneath the coil that dies out within 150 milliseconds without spreading, yielding a matrix of mostly zeros that compresses easily ($\mathrm{PCI} \approx 0.18$ –0.23). Under propofol or midazolam general anesthesia, a strong TMS pulse triggers a massive, stereotypical slow wave that sweeps uniformly across the cortex. Because every region does the exact same thing at the same time, the redundant matrix also compresses into a tiny dictionary ($\mathrm{PCI} \approx 0.12$ –0.30).

Only in a conscious brain does the magnetic perturbation propagate far across fronto-parietal and occipital networks while triggering different firing timings in each specialized region.

That combined integration and differentiation resists compression, yielding a high PCI between 0.44 and 0.67. In 2016, Silvia Casarotto, Marcello Massimini, and Steven Laureys validated an empirical cutoff of $\mathrm{PCI}^{*} = 0.31$ across 150 benchmark recordings in Milan and Liège. That single threshold separated 100 percent of conscious states—including dreaming REM sleep, ketamine dreams, and locked-in syndrome—from unconscious sleep and anesthesia. When they applied that $\mathrm{PCI}^{*} = 0.31$ threshold to thirty-eight behaviorally unresponsive vegetative-state patients, nine of the thirty-eight patients generated high-complexity cortical echoes above 0.31, identifying a hidden subgroup whose brains retained the integrated-and-differentiated architecture of wakefulness despite total sensory and motor isolation.

CHAPTER 10

Predictive Processing and Controlled Hallucination

How Top-Down Cortical Priors and Precision Weighting Construct the Bodily Self

Anatomists mapping the primate visual system in the late twentieth century uncovered a wiring ratio that baffled classical textbook neurology. Standard models treated perception as a bottom-up assembly line. In that view, retinal ganglion cells feed the lateral geniculate nucleus (LGN) of the thalamus, which sends signals forward to primary visual cortex V1, where simple cells detect edges and pass features up to V2, V4, and the inferior temporal lobe. Yet when neuroanatomists counted the actual synapses terminating on relay cells in the lateral geniculate nucleus, only about ten percent arrived from the retina. Fully thirty percent of the synapses in the LGN—and a vast majority of inter-areal fibers across the visual hierarchy—were top-down feedback projections descending *backward* from higher cortex to lower sensory relays.

Why would the brain wire far more cables running backward toward the eyes than running forward from them? The conceptual answer dates back to German physicist and physician Hermann von Helmholtz. In the 1867 third volume of his *Handbuch der Physiologischen Optik*, Helmholtz argued that retinal signals are inherently ambiguous and noisy. Locked inside the dark skull, the brain can never inspect external objects directly. Instead, perception operates as an automatic *unconscious inference* in which the mind guesses the hidden worldly causes most likely to have generated its sensory inputs.

In 1999, computational neuroscientists Rajesh Rao and Dana Ballard at the University of Rochester translated Helmholtz's insight into the circuit architecture of hierarchical predictive coding. Between 2005 and 2013, Karl Friston at University College London and Anil Seth at the University of Sussex generalized this model across cortical perception and bodily self-awareness. In predictive coding, deep pyramidal neurons in higher cortical areas continuously generate top-down predictions $g(\mu)$ of expected sensory input based on an internal hypothesis μ. Those descending fibers transmit predictions downward to suppress activity in lower sensory layers. Superficial pyramidal cells in the lower layers compare the top-down prediction against the actual bottom-up sensory signal s and transmit *only* the unexplained residual prediction error $\varepsilon_s = s - g(\mu)$ back up the hierarchy.

Crucially, not all prediction errors carry equal weight. When walking through a dark, foggy room, retinal signals have high variance σ_s^2, so the brain dials down sensory synaptic gain and relies heavily on its internal prior μ_{prior}. In bright daylight, high sensory precision $\Pi_s = 1/\sigma_s^2$ amplifies bottom-up errors to override the prior. Updating the brain's conscious perceptual state μ follows precision-weighted Bayesian inference and gradient descent on variational free energy:

$$\dot{\mu} = D\mu - \Pi_\mu \varepsilon_\mu + \left(\frac{\partial g}{\partial \mu}\right)^T \Pi_s \varepsilon_s, \qquad \mu_{\text{post}} = \frac{\Pi_\mu \mu_{\text{prior}} + \Pi_s s}{\Pi_\mu + \Pi_s}$$

As Anil Seth synthesized this framework, conscious perception is a *controlled hallucination*. What we consciously experience is not the bottom-up error signal itself, but the brain's top-down generative prediction μ_{post} kept on a tight leash by sensory prediction errors. Even the intimate feeling that our physical body belongs

to us is a malleable Bayesian inference. In 1998, psychiatrists Matthew Botvinick and Jonathan Cohen at the University of Pittsburgh seated healthy volunteers with their real left arm hidden behind a screen while a lifelike rubber left hand rested on the table in front of them. Because visual spatial precision Π_{vis} exceeds proprioceptive joint precision Π_{prop}, stroking both hands synchronously with two paintbrushes forced the brain's multisensory model to bind the felt touch to the visible rubber hand. Within ten minutes of synchronous brush strokes in Botvinick and Cohen's Pittsburgh trials, subjects misjudged the physical location of their own unseen index finger by up to six centimeters toward the rubber prosthesis, while asynchronous brushing produced zero illusion and zero proprioceptive drift.

CHAPTER 11

Higher-Order Metacognition and Prefrontal Monitoring

How Anterior Prefrontal Lesions Blind Introspection While Leaving Vision Intact

Why can a blindsight patient point accurately to a moving grating in their blind field at ninety percent accuracy while sincerely insisting that they see nothing at all? Philosopher David Rosenthal at the City University of New York formulated Higher-Order Thought (HOT) theory between 1986 and 2005 to explain this dissociation. According to Rosenthal, first-order sensory representations in the posterior visual cortex are never conscious on their own. A neural representation of a vertical grating in area V1 or MT remains completely unconscious unless a higher-order monitoring circuit in the prefrontal cortex forms a meta-representation asserting that the subject is currently seeing that grating.

For years, critics dismissed Higher-Order theories on empirical grounds. Whenever neuroscientists compared seen versus unseen visual stimuli in the laboratory, seen stimuli almost always produced higher objective task accuracy (d', or signal-detection sensitivity) than unseen stimuli. When a subject performs better at a task while the prefrontal cortex is active, an obvious confound arises. Is the prefrontal cortex generating conscious visual experience itself, or is it merely driving attention, working memory, and button-press performance?

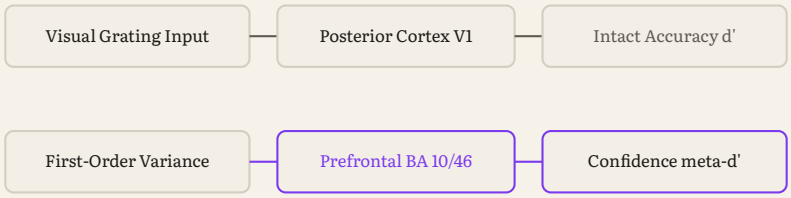


FIG. 11.1 — Matched first-order sensitivity d′ in visual cortex (top) dissociated from prefrontal metacognitive confidence meta-d′ (bottom).

In 2006, cognitive neuroscientists Hakwan Lau and Richard Passingham at the University of Oxford broke that stalemate by exploiting a U-shaped property of visual metacontrast masking. When a diamond or square target is followed by a non-overlapping surrounding mask, objective discrimination accuracy traces a U-shaped curve across time delays. Lau and Passingham identified two distinct mask delays—thirty-three milliseconds and 104 milliseconds—that yielded the exact same objective forced-choice accuracy (d'). Despite identical discrimination performance, subjects reported consciously seeing the target shape significantly more often at the 104-millisecond delay. Inside the Oxford fMRI scanner, posterior visual cortex activity remained flat between the two performance-matched conditions. By contrast, a localized region in the left mid-dorsolateral prefrontal cortex (Brodmann area 46) tracked the jump in subjective visual awareness.

To formalize this separation between first-order perceptual sensitivity and second-order introspection, Brian Maniscalco and Hakwan Lau introduced the signal-detection metric $meta\text{-}d'$ in 2012. Standard d' measures how well an observer's primary button presses discriminate stimulus S_1 from S_2. The metric $meta\text{-}d'$, by contrast, calculates what first-order sensitivity an ideal Bayesian

observer would need in order to produce the subject's trial-by-trial confidence ratings. Dividing the two quantities yields the metacognitive efficiency ratio $M = meta\text{-}d'/d'$, which isolates introspective monitoring accuracy from raw visual sharpness. Two years earlier, in 2010, Elisabeth Rounis and Hakwan Lau had demonstrated that applying continuous theta-burst transcranial magnetic stimulation over bilateral dorsolateral prefrontal cortex in healthy volunteers selectively depressed metacognitive visibility ratings without lowering objective discrimination sensitivity d'.

In 2010 and 2014, neuroscientist Stephen Fleming and his colleagues at University College London and New York University used a continuous psychophysical staircase to clamp every participant's first-order visual discrimination accuracy at seventy-one percent correct. By dynamically adjusting grating contrast from trial to trial, the staircase eliminated any performance advantage between subjects. Among healthy volunteers in the 2010 *Science* study, individual differences in $meta\text{-}d'$ varied widely and correlated specifically with gray-matter volume and white-matter microstructure in the anterior prefrontal cortex (rostrolateral Brodmann area 10). When Fleming's team tested neurosurgical patients with localized resections of the anterior prefrontal cortex in 2014, those patients discriminated two-patch visual contrast grids at the exact same seventy-one percent staircase accuracy as healthy controls and temporal-lobe patients. Yet their perceptual metacognitive efficiency ratio ($meta\text{-}d'/d'$) dropped by nearly forty percent, leaving their confidence ratings disconnected from their actual visual accuracy.

CHAPTER 12

The Adversarial Testing of Rival Theories

What Happened When Global Workspace and Integrated Information Theory Faced the Same Data

At a consciousness conference in Bremen, Germany, in June 1998, neuroscientist Christof Koch made a public wager with philosopher David Chalmers. Inspired by rapid advances in single-unit primate electrophysiology and human functional neuroimaging, Koch bet a case of fine wine that within twenty-five years—by June 2023—science would discover a clear, universally accepted neural signature of consciousness in the brain. Chalmers, skeptical that empirical correlation alone would bridge the theoretical divide, took the opposing side.

As the twenty-five-year deadline approached, the field had not converged on a single mechanism. Instead, consciousness neuroscience had split into rival theoretical camps. On one side stood Stanislas Dehaene's Global Neuronal Workspace Theory (GNWT), which locates conscious broadcast in prefrontal-parietal ignition. On the other side stood Giulio Tononi's Integrated Information Theory (IIT), which locates phenomenal experience in a posterior cortical *hot zone* spanning visual and temporal areas. Worse, because each laboratory designed its own behavioral tasks and statistical pipelines, advocates could always explain away awkward data after the fact.

In 2019, neuroscientists Lucia Melloni, Liad Mudrik, and Michael Pitts organized the Cogitate Consortium—an open-science adversarial collaboration funded by the Templeton World Charity Foundation—to force Dehaene and Tononi into a pre-registered head-to-head test. Before a single volunteer was scanned, both

theorists signed off on an identical experimental protocol. Across six independent laboratories, researchers presented supraliminal faces, objects, letters, and false-font symbols for exact durations of 0.5, 1.0, and 1.5 seconds to 256 human participants (120 in fMRI, 102 in magnetoencephalography, and 34 epilepsy patients with intracranial cortical electrodes), locking in three decisive tests:

- **Stimulus Maintenance Duration (0.5 s vs. 1.0 s vs. 1.5 s)**: IIT predicted sustained neural firing and decoding across the posterior visual hot zone for the entire duration an image stays consciously visible. GNWT, by contrast, predicted a brief onset ignition burst (~300 ms) followed by silent synaptic maintenance until a second ignition burst at stimulus offset.
- **Prefrontal Content Decoding (Category vs. Orientation)**: GNWT predicted that the prefrontal workspace must decode every consciously perceived attribute, including both object category and grating orientation. IIT predicted that orientation can remain conscious inside posterior visual cortex without prefrontal representation.
- **Inter-Areal Network Synchrony**: IIT required sustained gamma- and beta-band phase synchronization linking early visual cortex (V1/V2) with ventral temporal fusiform areas throughout the conscious percept. GNWT instead required long-range fronto-parietal synchrony during workspace ignition.

When the Cogitate Consortium unveiled its results in June 2023 and published the full peer-reviewed dataset in *Nature* in April 2025, the pre-registered ledger dealt sharp blows to both theories. Posterior visual areas faithfully tracked the full 0.5-, 1.0-, and 1.5-second duration of conscious percepts and decoded stimulus orientation as IIT predicted. However, the intracranial recordings failed to find the sustained posterior inter-areal syn-

chronization that IIT required for causal integration. Conversely, the prefrontal cortex strongly decoded stimulus category at onset as GNWT predicted, but failed to decode conscious stimulus orientation and lacked the predicted second ignition burst when the stimulus turned off. Neither workspace broadcast nor posterior integration survived the adversarial test unscathed. At the twenty-sixth annual meeting of the Association for the Scientific Study of Consciousness at New York University in June 2023, the Cogitate Consortium presented its three-modality results across 256 human subjects. Immediately afterward, Christof Koch walked onto the auditorium stage and handed David Chalmers a bottle of 1978 Portuguese Madeira wine to settle their twenty-five-year wager.